

# IMAGE PROCESSING & ANALYSIS

Post Processing Steps

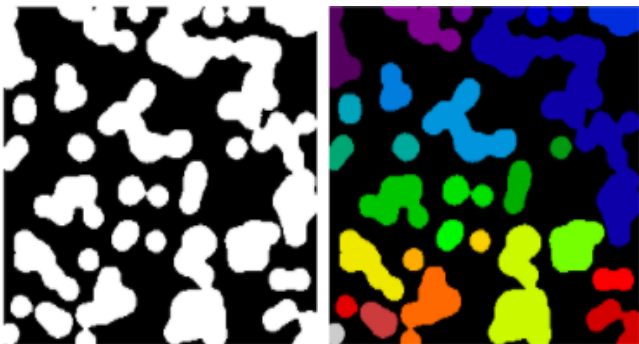
- After segmenting an image, we usually want to:
  - Clean up  
the binary image or  
each one of the resulting regions
  - Know something  
about the regions found ...
    - ◆ How many objects are in the image?
    - ◆ What the the area, perimeter, position, ...  
of the distinct object components ?

## Post-processing steps

- Morphologic operations, for
  - closing gaps and holes
  - removing “noses”
  - computing centerlines
  - ...
- Extraction of connected components (BLOBs)
- Quantitative analysis
  - center, area, perimeter, volume, ...
  - principal axes (for orientation)
  - ...
- ...

# Computer Extraction of connected components (BLOBs)

- Typically, the segmented image contains multiple objects that should be returned individually.
- Typically, the objects we are interested in are characterized by forming a connected set of pixels.
- To obtain the individual regions we must compute the connected components of the segmented region.
  - => to define when two pixels should be considered connected.
- On a rectangular pixel grid, there are 2 possible definitions of connectivity:
  - 4-connectivity: two pixels are considered part of the same segment if they are horizontally or vertically adjacent
  - 8-connectivity: two pixels are considered part of the same segment if they are horizontally, vertically, or diagonally adjacent
- A **component/region labelling algorithm** determines all the connected components of an image and attributes a unique label to all the points of the same component/region.
- Algorithms:
  - recursive
  - sequential



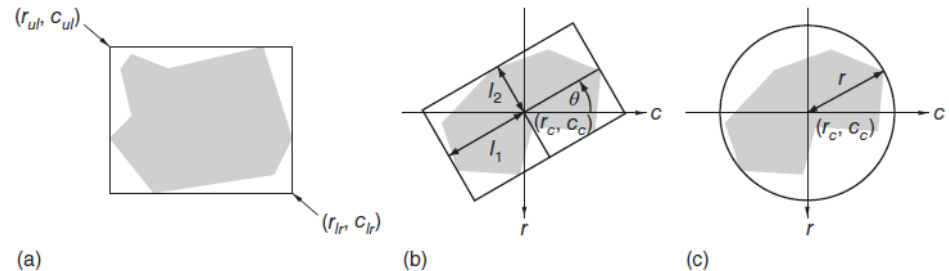
|   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 1 | 0 | 0 | 0 | 1 | 1 | 1 |
| 1 | 1 | 1 | 0 | 0 | 1 | 0 |
| 1 | 1 | 0 | 0 | 0 | 1 | 0 |
| 0 | 0 | 0 | 1 | 0 | 1 | 0 |
| 0 | 0 | 1 | 1 | 0 | 0 | 0 |
| 0 | 0 | 0 | 0 | 0 | 1 | 1 |

|   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 1 | 0 | 0 | 0 | 2 | 2 | 2 |
| 1 | 1 | 1 | 0 | 0 | 2 | 0 |
| 1 | 1 | 0 | 0 | 0 | 2 | 0 |
| 0 | 0 | 0 | 3 | 0 | 2 | 0 |
| 0 | 0 | 3 | 3 | 0 | 0 | 0 |
| 0 | 0 | 0 | 0 | 0 | 4 | 4 |

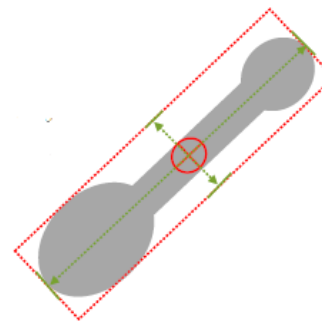
- In most applications it is required that we determine one or more characteristic quantities from the segmented regions or contours.
- The quantities we determine are called features; typically they are real numbers.
- The process of determining the features is called feature extraction (\*).
- There are different kinds of features:
  - Region and contour features that can be extracted from binary or graylevel images.
- Binary shape analysis:
  - While binary images contain much less information than their grayscale counterparts, they embody shape and size information that is highly relevant for object recognition.
  - Features: area, perimeter, circularity, aspect ratio, moments, ... (see next slide)
- Gray/Color value features:
  - Use the gray/color values in the image within the region.
  - Features: min, max, mean, standard deviation, histogram(s), moments, ...

(\*) - Do not confuse with "feature point extraction/detection"  
( = detection of points that can be differentiated from its neighboring image points)

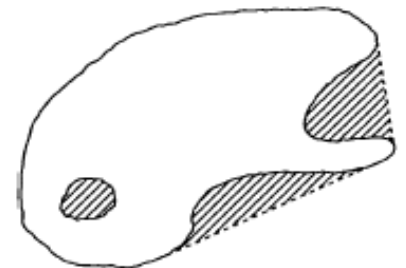
- There are many simple tests of shape that can be made to confirm the identity of objects or to check for items such as defects.
- These include measurements of
  - area and perimeter
  - bounding box, length, width, angle
  - circularity, elongation, aspect ratio
  - length of maximum dimension
  - moments relative to the centroid
    - ◆ note: some of the statistics above can be obtained from centralized moments
  - number and area of holes
  - area and dimensions of the convex hull and enclosing rectangle
  - number of sharp corners
  - numbers and types of skeleton nodes
  - ...



- (a) Smallest axis-parallel enclosing rectangle of a region.  
 (b) Smallest enclosing rectangle of arbitrary orientation.  
 (c) Smallest enclosing circle.

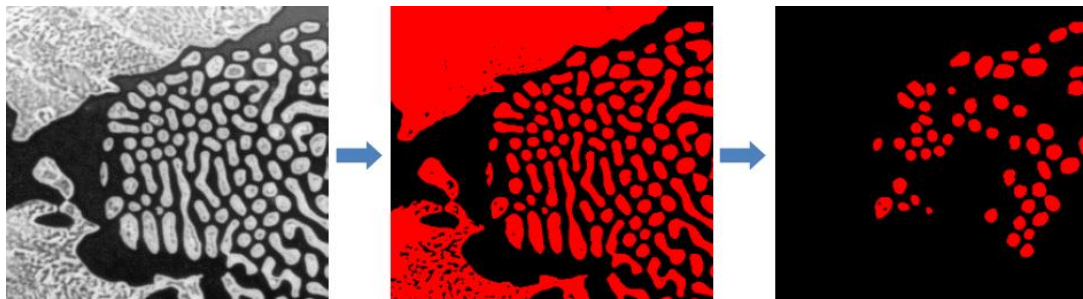


Enclosing rectangle,  
centroid and main axis

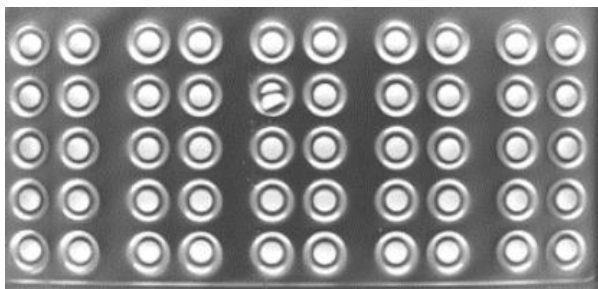


Convex hull and convex deficiency.

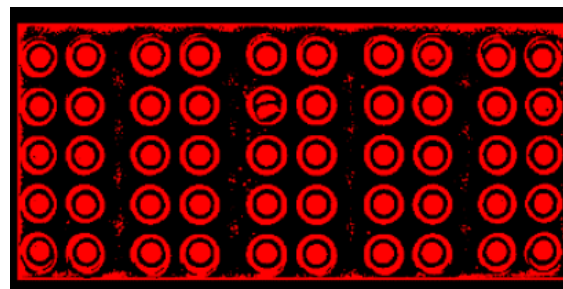
- Keeps or removes particles based on geometric features
  - area, width, height, aspect ratio and other features are commonly used to filter
- Typically used on binary images
- Cleans up noisy images



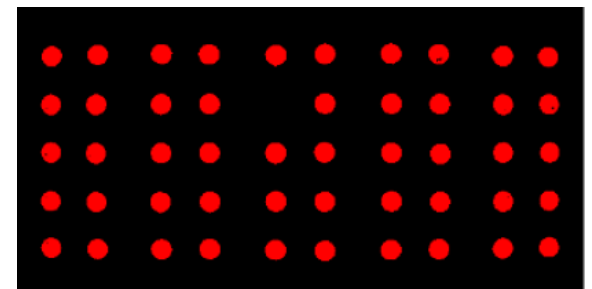
Particles with circularity factor  $< 1.1$



Original



Threshold



Particle filter

## Structural Analysis and Shape Descriptors

- **cv::connectedComponents**
  - computes the connected components labeled image of boolean image
- **cv::connectedComponentsWithStats**
  - computes the connected components labeled image of boolean image and also produces a statistics output for each label
- **cv::moments**
  - Calculates all of the moments up to the third order of a polygon or rasterized shape
- **cv::HuMoments**
  - Calculates seven Hu invariants
- **cv::findContours**
  - Finds contours in a binary image
- **cv::arcLength**
  - Calculates a contour perimeter or a curve length
- **cv::contourArea**
  - Calculates a contour area
- **cv::boundingRect**
  - Calculates the up-right bounding rectangle of a point set or non-zero pixels of gray-scale image
- **cv::minAreaRect**
  - Finds a rotated rectangle of the minimum area enclosing the input 2D point set
- **cv::minEnclosingCircle**
  - Finds a circle of the minimum area enclosing a 2D point set
- **cv::isContourConvex**
  - Tests a contour convexity
- **cv::convexHull**
  - Finds the convex hull of a point set
- ... (and several other)



# **IMAGE PROCESSING & ANALYSIS**

Morphological image processing

- Segmentation results often contain unwanted noisy parts. Furthermore, sometimes the segmentation will contain parts in which the shape of the object we are interested in has been disturbed, for example, because of noise.
- Therefore, often we need to modify the shape of the segmented regions to obtain the desired results.
- This is the subject of the field of **mathematical morphology**, which can be defined as a theory for the analysis of spatial structures.
- Mathematical morphology provides a set of extremely useful operations that enable us not only to modify the shape of objects but also to smooth images or enhance edges (*see next slides*)
- **Morphological image processing**
  - a set of operations by which the spatial structure of objects within an image is modified.
- **Morphological operations** can be defined on **binary** and **gray value images**.
- **Operations (binary / grayscale):**

|                                                                                                                                                          |                                                                                                                                                                                                                                                                       |
|----------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>• Dilation</li><li>• Erosion</li><li>• Opening = Dilation(Erosion)</li><li>• Closing = Erosion(Dilation)</li></ul> | <b>Operations (grayscale):</b> <ul style="list-style-type: none"><li>• Morphological gradient</li><li>• TopHat - reveals areas that are lighter than the surrounding region</li><li>• BottomHat - reveals areas that are darker than the surrounding region</li></ul> |
|----------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
- Two fundamental parameters of those operations are the **shape** and **center** of the **structuring element**.



Original image



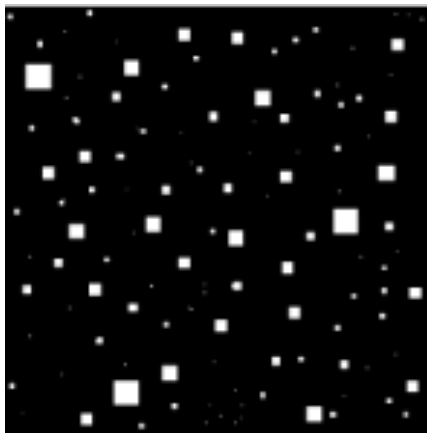
Image after  
morphological processing



Original image



Image after  
morphological processing  
(opening)



Original image

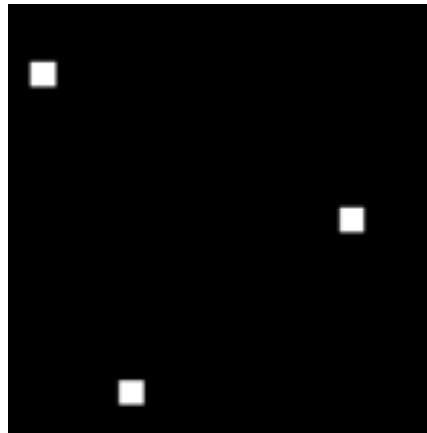


Image after  
morphological processing



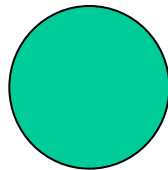
Image after  
morphological processing  
(closing)

- The arguments to dilation and erosion are:
  - a binary image **B**
  - a structuring element **S**
- **Structuring element (or kernel)**
  - A structuring element is a shape mask used in the basic morphological operations.
  - They can be any shape and size that is digitally representable, and each has an origin.
  - It can even correspond to a particular shape that is being sought for in the image.

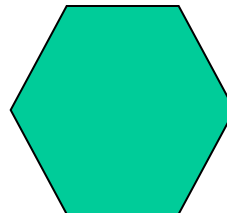
box



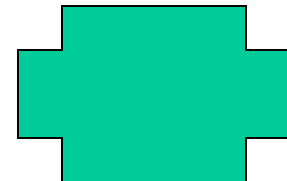
disk



hexagon

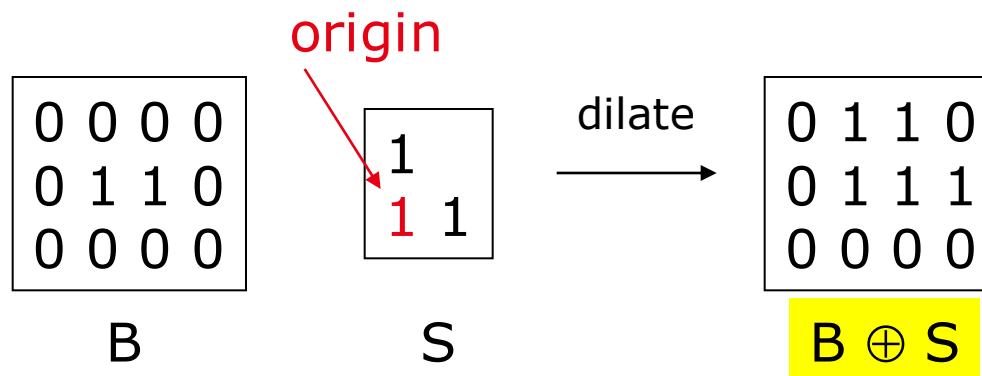


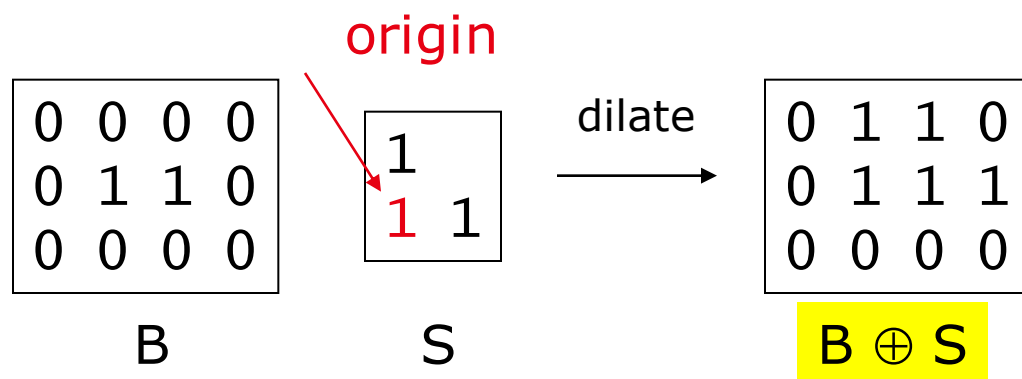
something



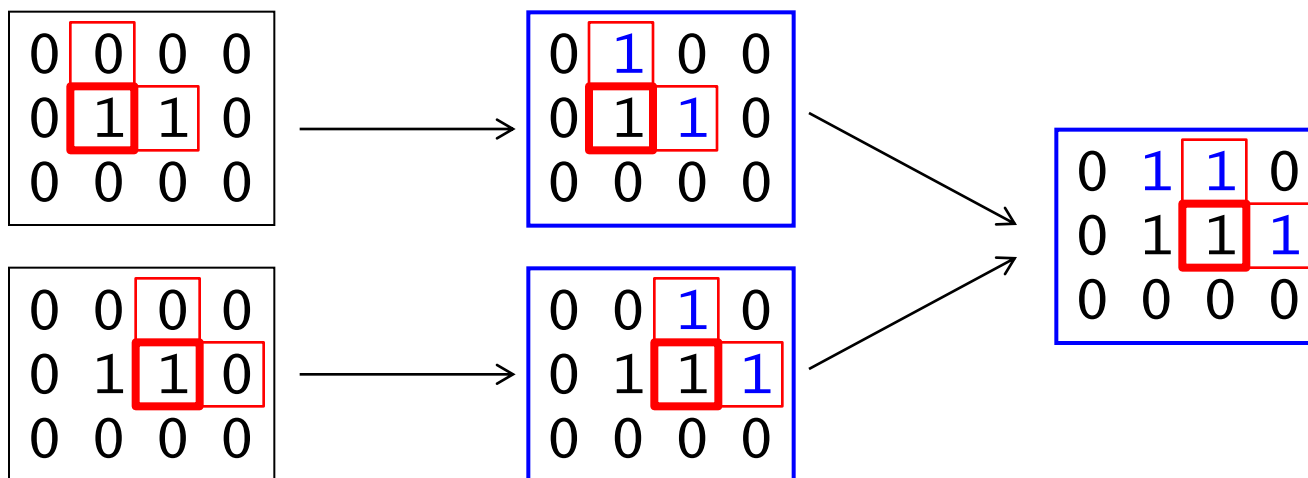
## **dilate(B,S)**

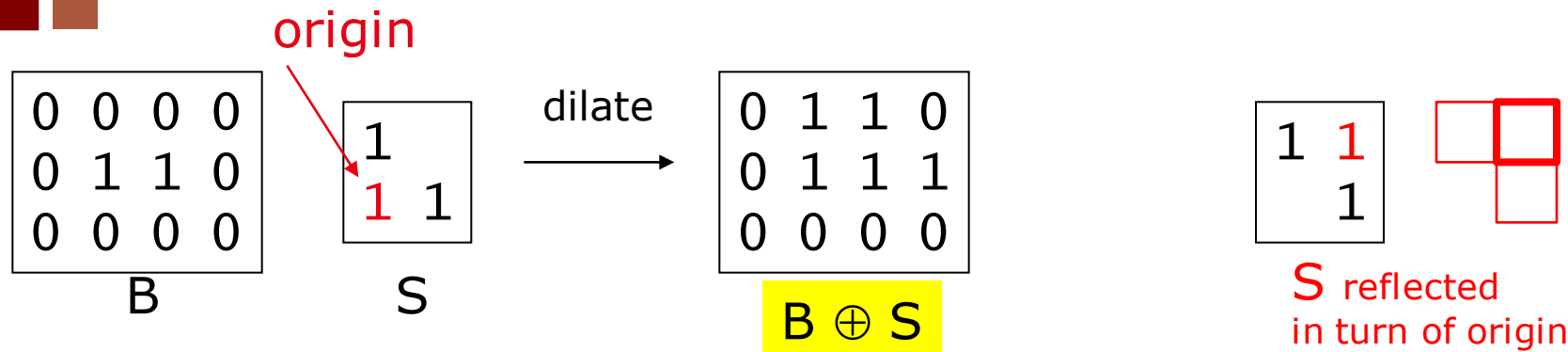
- takes binary image B,
  - places the origin of structuring element S over each 1-pixel, and
  - ORs the structuring element S into the output image at the corresponding position.
- 
- Dilation is both associative and commutative
    - this fact allows breaking a complex shape into several simpler shapes which can be combined as a sequence of dilations



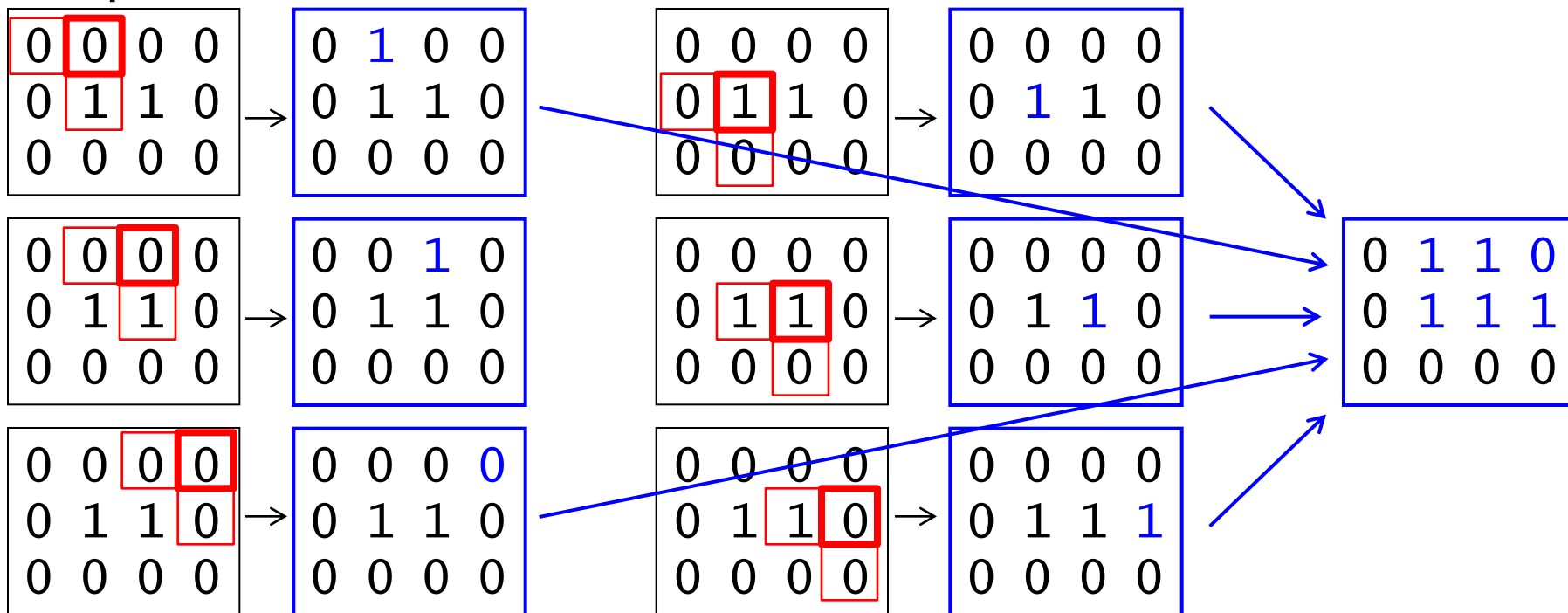


Steps:



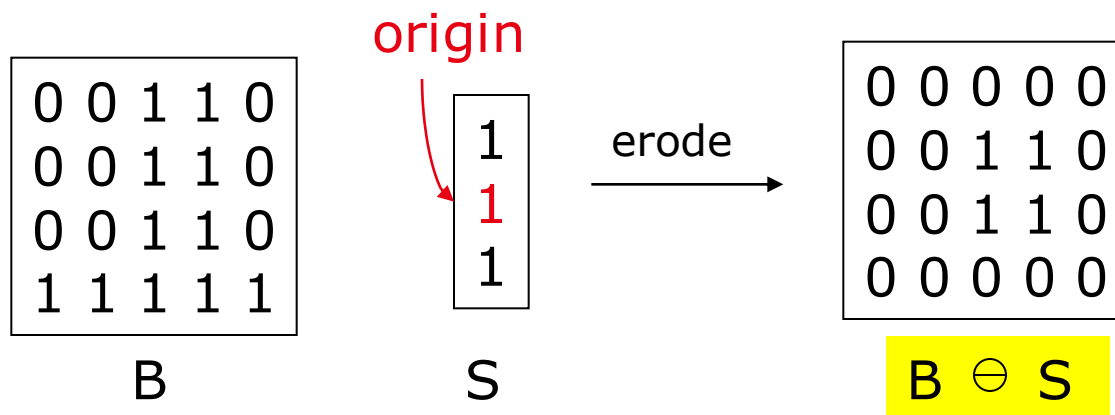


**Steps** (alternative formulation):  $\text{dilate}(B, S)(x, y) = \max \{B(x-x', y-y') \mid (x', y') \in S\}$

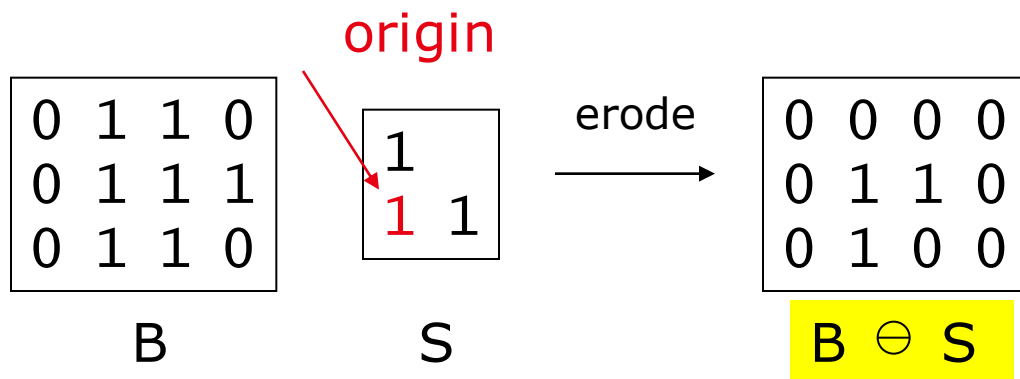


**erode(B,S)**

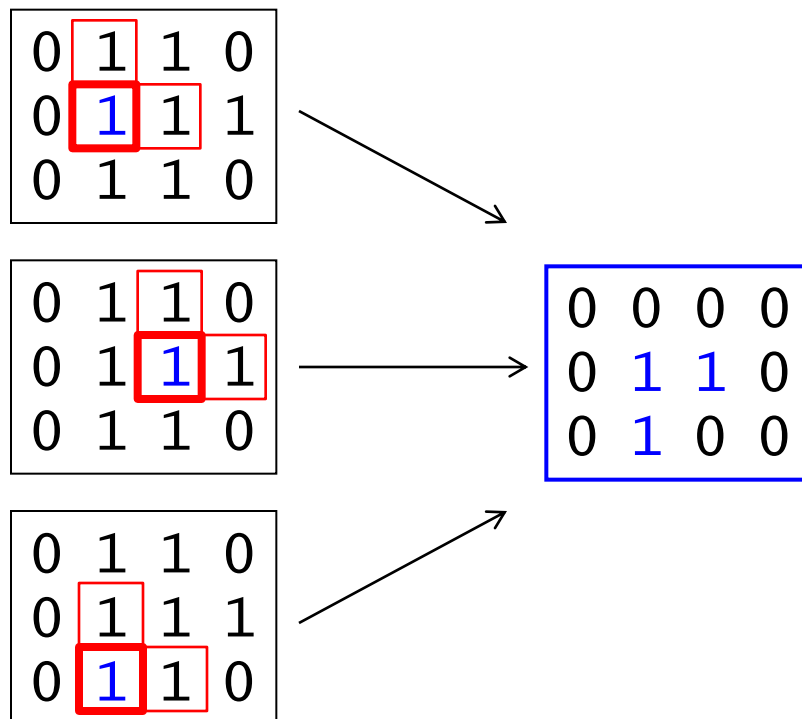
- takes a binary image B,
  - places the origin of structuring element S over every pixel position, and
  - ORs a binary 1 into that position of the output image only if every position of S (with a 1) covers a 1 in B.
- Note:  
erosion gives all the locations where the structuring element is contained in the image

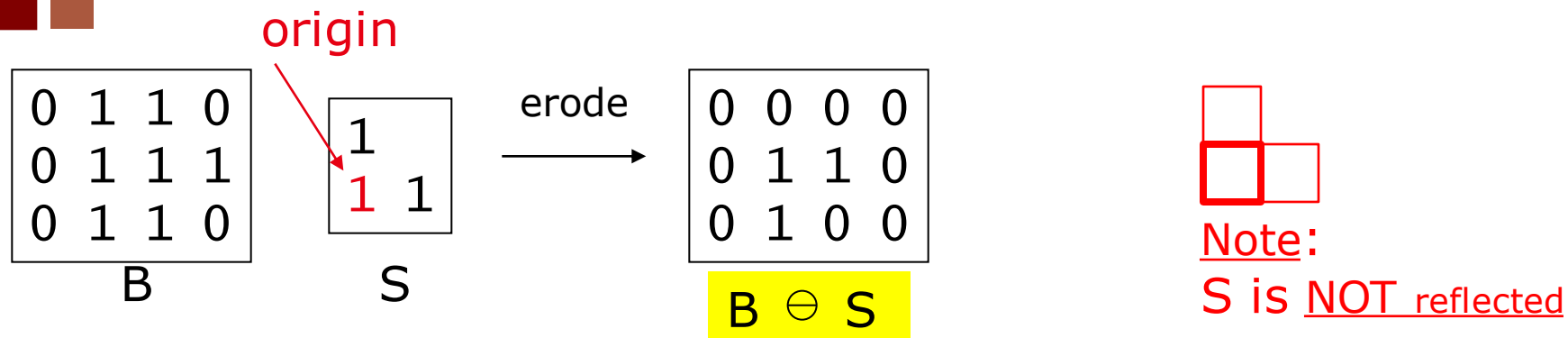




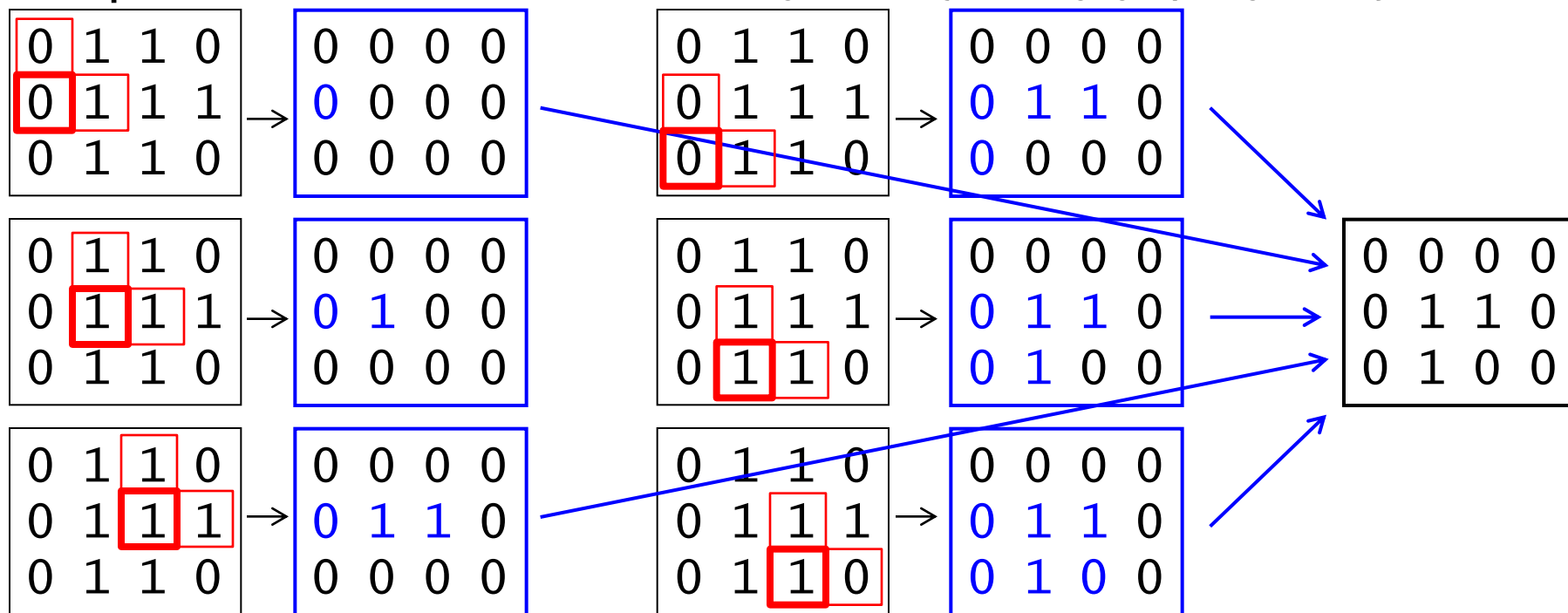


Steps:





Steps (alternative formulation):  $\text{erode}(B, S)(x, y) = \min \{B(x+x', y+y') \mid (x', y') \in S\}$



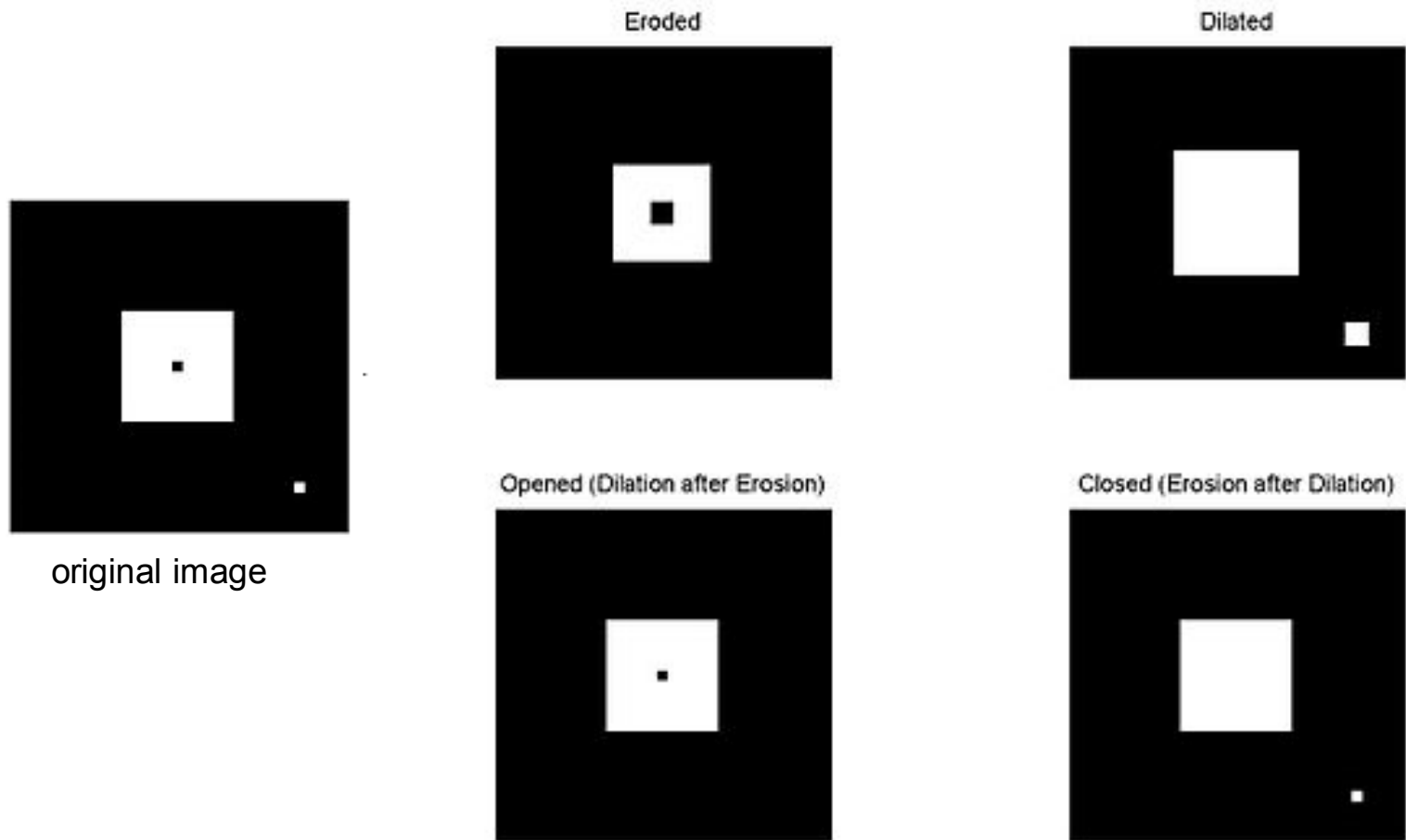
- Dilation
  - expands the connected sets of 1s of a binary image.
  - it can be used for
    - ◆ growing features
    - ◆ filling holes and gaps
- Erosion
  - shrinks the connected sets of 1s of a binary image.
  - it can be used for
    - ◆ shrinking features
    - ◆ removing bridges, branches and small protrusions
- Dilation and erosion can be useful in a wide variety of contexts such as:
  - removing noise,
  - isolating individual elements, and
  - joining disparate elements in an image.
- Morphological operators can also be used
  - to find intensity bumps or holes in an image and
  - to calculate image gradients.

### Opening

- is the compound operation of erosion followed by dilation (with the same structuring element)
- **$\text{open}(\mathbf{B}, \mathbf{S}) = \text{dilate}(\text{erode}(\mathbf{B}, \mathbf{S}), \mathbf{S})$**

### Closing

- is the compound operation of dilation followed by erosion (with the same structuring element)
- **$\text{close}(\mathbf{B}, \mathbf{S}) = \text{erode}(\text{dilate}(\mathbf{B}, \mathbf{S}), \mathbf{S})$**



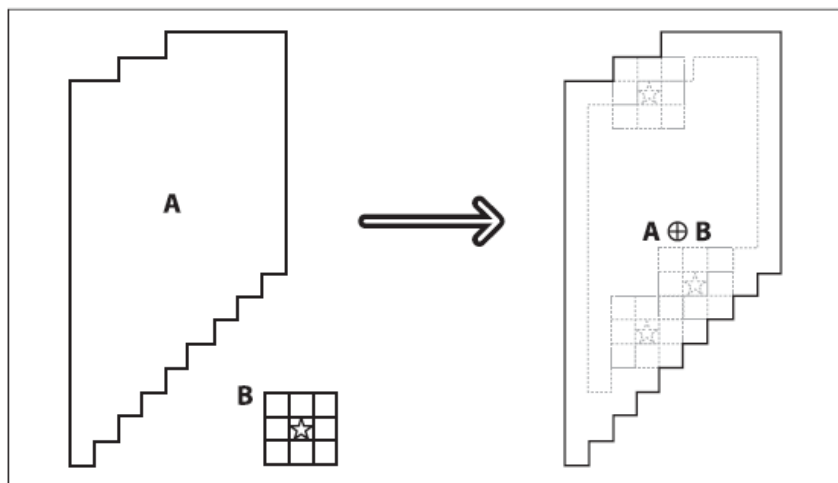
source:

[https://en.wikibooks.org/wiki/Sensory\\_Systems/Computer\\_Models/Descriptive\\_Simulations\\_of\\_Visual\\_Information\\_Processing](https://en.wikibooks.org/wiki/Sensory_Systems/Computer_Models/Descriptive_Simulations_of_Visual_Information_Processing)

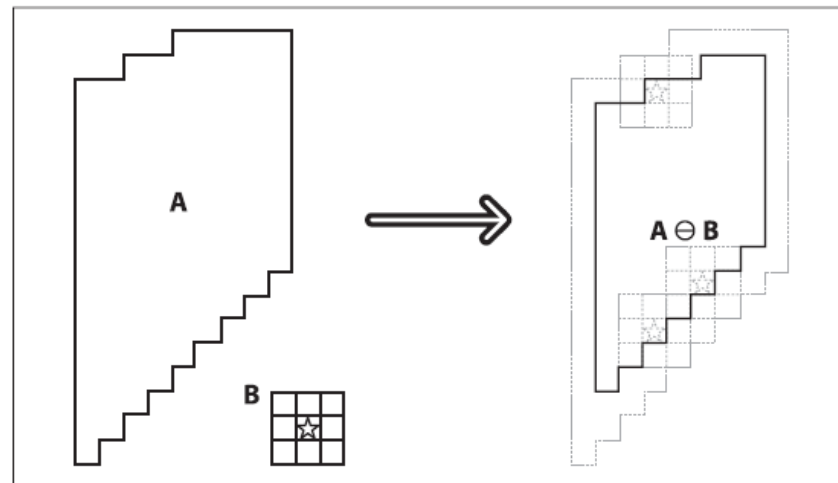
- Dilation
  - $(g \oplus s)(x,y) = \max \{g(x-x',y-y') + s(x',y') | (x',y') \in D_s\}$
- Erosion
  - $(g \ominus s)(x,y) = \min \{g(x+x',y+y') - s(x',y') | (x',y') \in D_s\}$
- being
  - $g(x,y)$  a graylevel image
  - $s(x,y)$  a structuring element (graylevel)
    - ◆  $D_s$  a binary matrix that defines the locations in the neighborhood that must be included in the max/min operations
- When the structuring element is flat one can define  $s(x',y') = 0$  for  $(x',y') \in D_s$
- So ...
  - Dilation
    - ◆  $(g \oplus s)(x,y) = \max \{g(x-x',y-y') | (x',y') \in D_s\}$
  - Erosion
    - ◆  $(g \ominus s)(x,y) = \min \{g(x+x',y+y') | (x',y') \in D_s\}$
- ...and if the structuring element is symmetric ...
  - Dilation
    - ◆  $(g \oplus s)(x,y) = \max \{g(x+x',y+y') | (x',y') \in D_s\}$
  - Erosion
    - ◆  $(g \ominus s)(x,y) = \min \{g(x+x',y+y') | (x',y') \in D_s\}$

$$\text{erode}(x, y) = \min_{(x', y') \in \text{kernel}} \text{src}(x + x', y + y')$$
$$\text{dilate}(x, y) = \max_{(x', y') \in \text{kernel}} \text{src}(x + x', y + y')$$

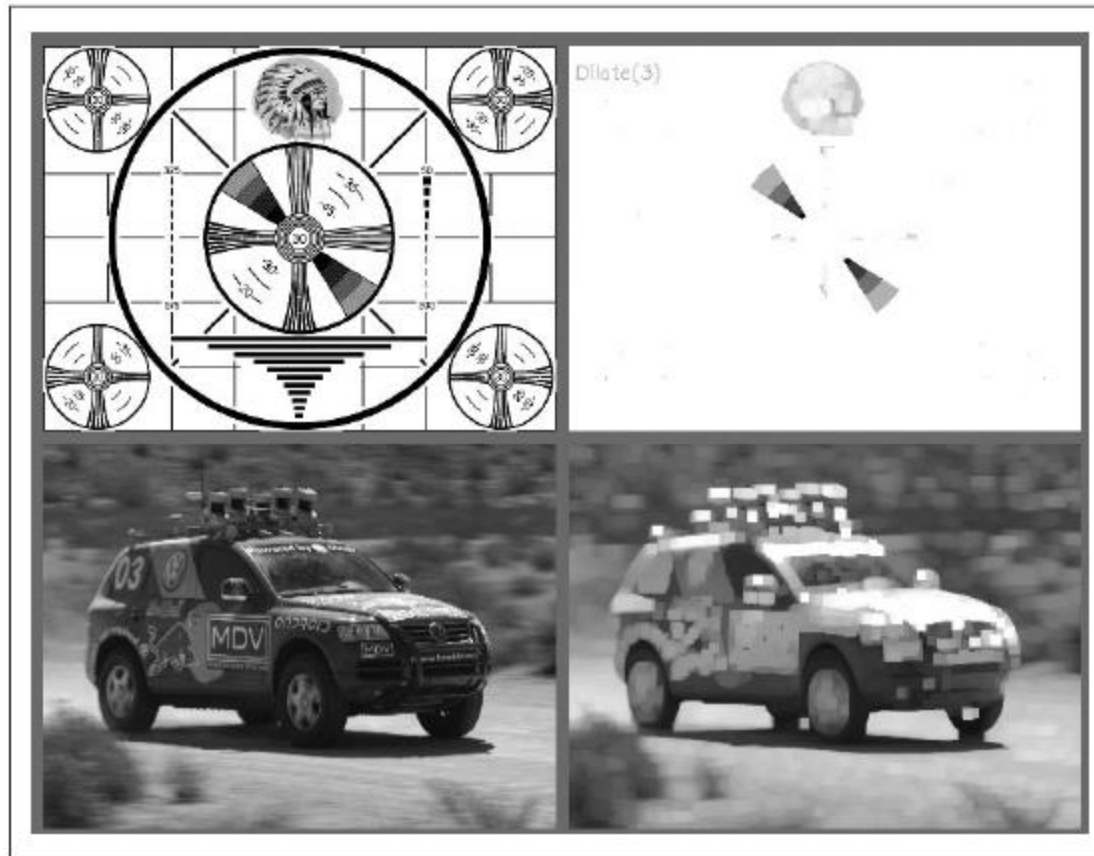
kernel  $\Leftrightarrow$   
structuring element



**Dilation:** take the **maximum** under the kernel B



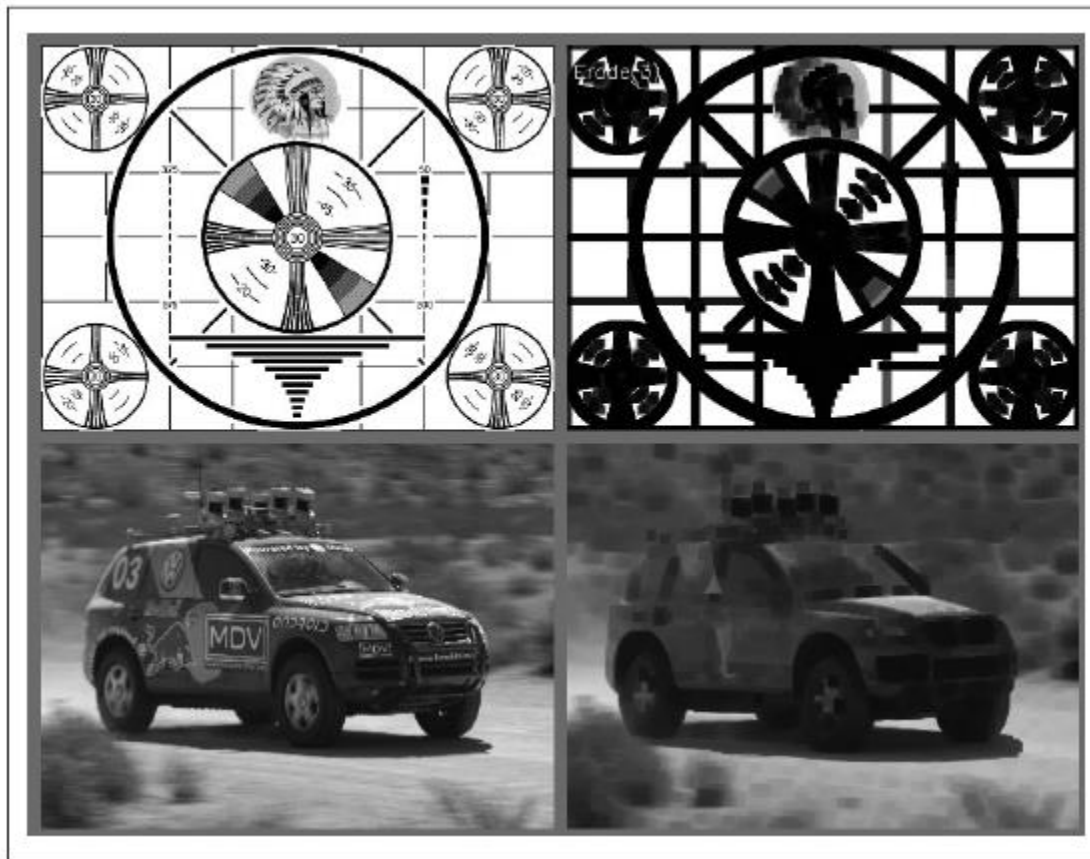
**Erosion:** take the **minimum** under the kernel B



Results of the dilation, or “max”, operator:  
bright regions are expanded and often joined  
(source: *Bradski book*)

(source: *Bradski book*)





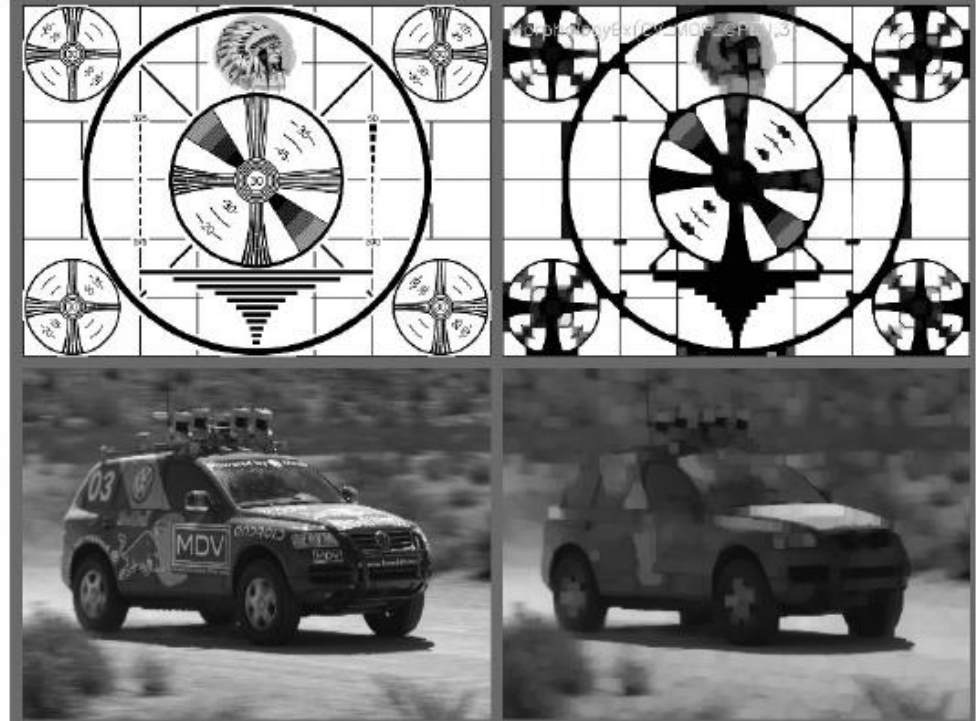
Results of the erosion, or “min”, operator:  
bright regions are isolated and shrunk  
(source: *Bradski book*)

(source: *Bradski book*)

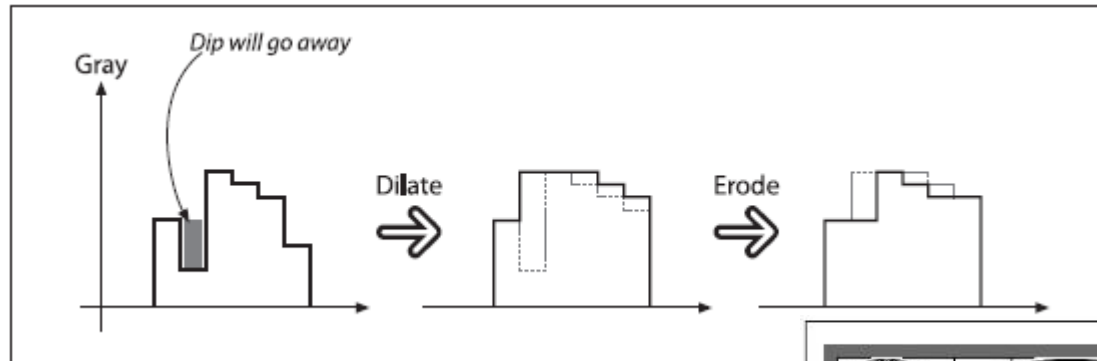


Morphological opening:  
the upward outliers  
are eliminated as a result

Results of opening on an image:  
small bright regions are removed,  
the remaining bright regions  
are isolated but retain their size  
(source: Bradski book)

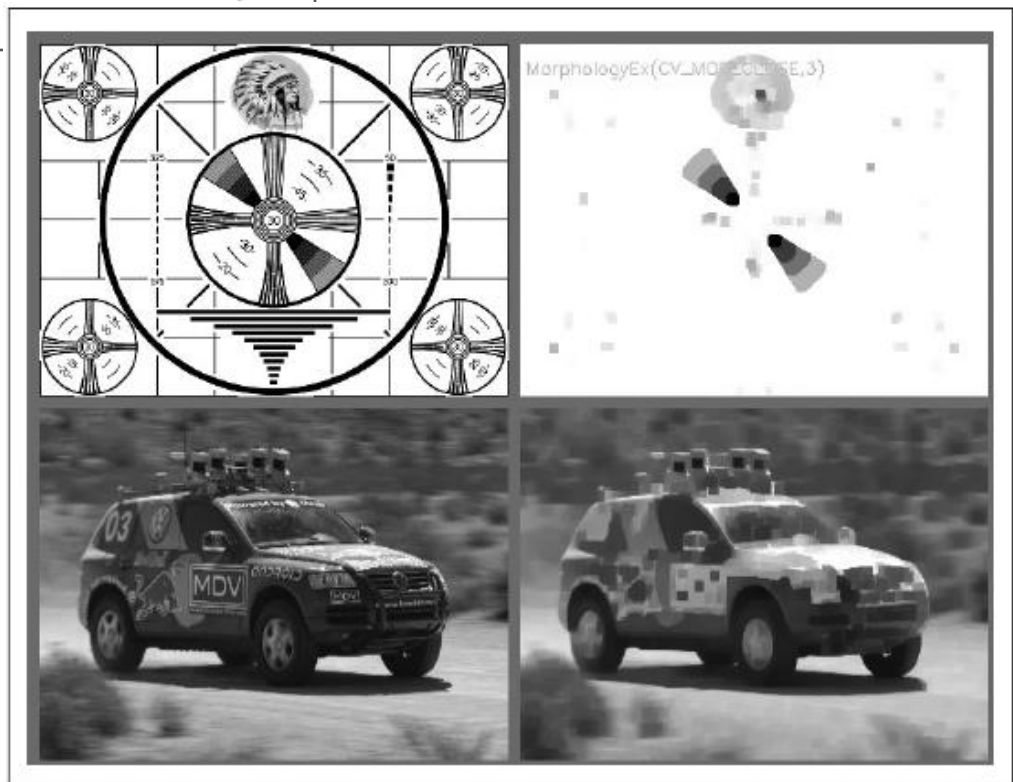


(source: Bradski book)



Morphological closing:  
the downward outliers  
are eliminated as a result

Results of closing on an image:  
bright regions are joined  
but retain their basic size  
(source: Bradski book)

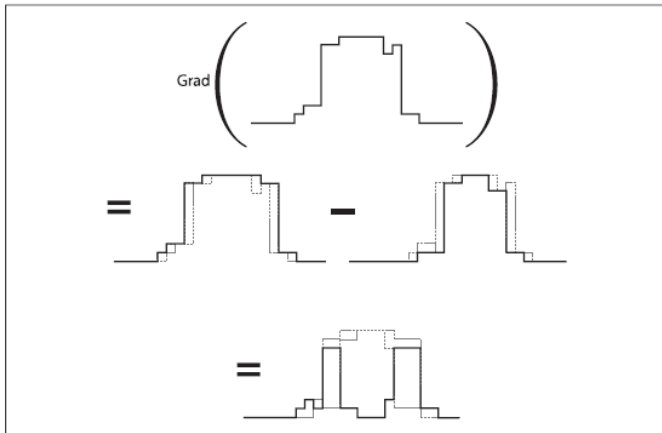


(source: Bradski book)

- Morphological gradient
  - **$\text{gradient}(\text{src}) = \text{dilate}(\text{src}) - \text{erode}(\text{src})$** 
    - ◆ the effect of this operation on a binary image would be simply to isolate perimeters of existing blobs.
- Top-hat <sup>\*</sup>
  - **$\text{TopHat}(\text{src}) = \text{src} - \text{open}(\text{src})$** 
    - ◆ reveals areas that are lighter than the surrounding region
- Black-hat <sup>\*\*</sup>
  - **$\text{BlackHat}(\text{src}) = \text{close}(\text{src}) - \text{src}$** 
    - ◆ reveals areas that are darker than the surrounding region

<sup>\*</sup>, <sup>\*\*</sup> - Bradski book ("Learning OpenCV") designations;  
"Black-hat" is also known as "Bottom-hat"

## Morphological gradient

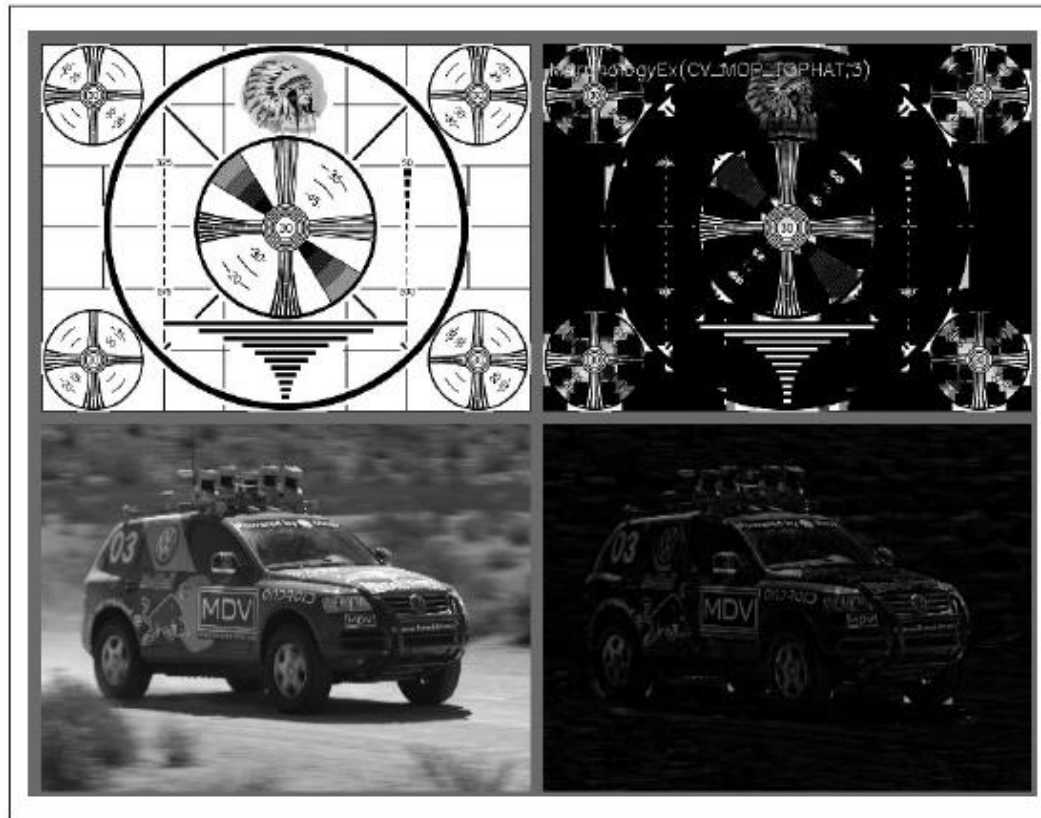


Morphological gradient applied to a grayscale image: as expected, the operator has its highest values where the grayscale image is changing most rapidly



Results of the morphological gradient operator: bright perimeter edges are identified

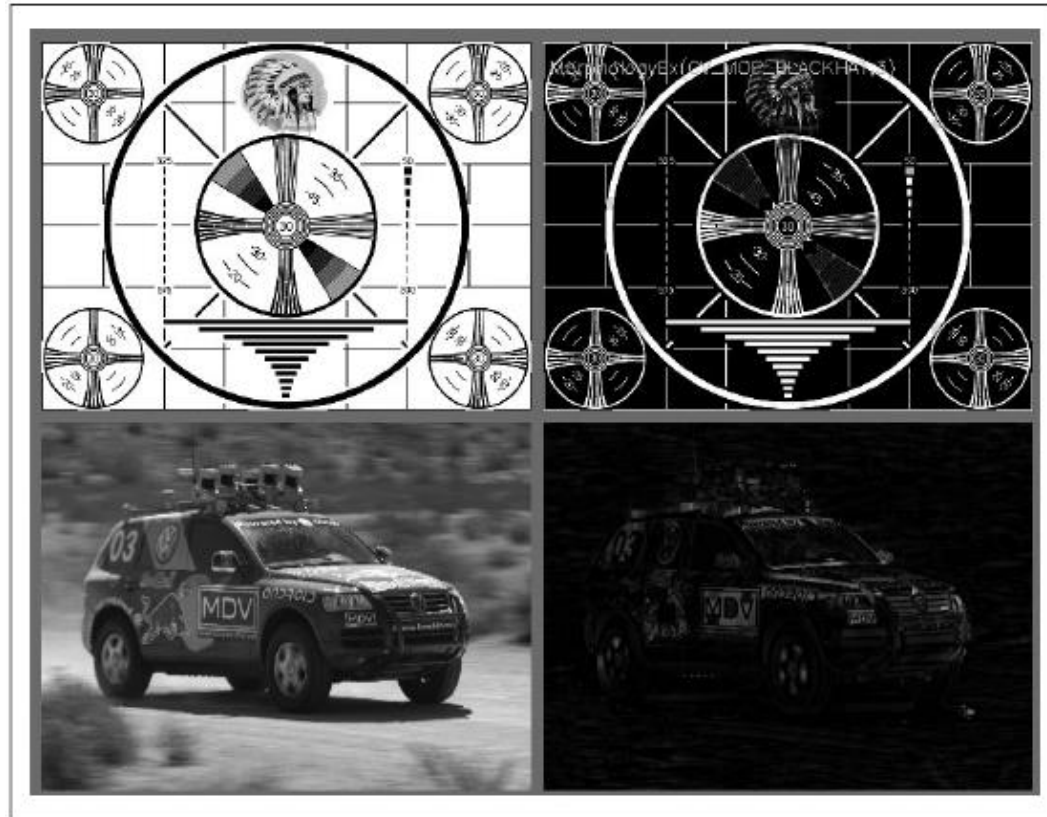
## Top-hat



Results of Top-Hat:  
bright local peaks are isolated



## Black-hat



Results of Black-Hat operation:  
dark holes are isolated

Image I



Erosion  $I \ominus B$



Dilation  $I \oplus B$



Opening  $I \circ B = (I \ominus B) \oplus B$



Closing  $I \bullet B = (I \oplus B) \ominus B$



Gradient  $I = (I \oplus B) - (I \ominus B)$



TopHat  $I = I - (I \ominus B)$



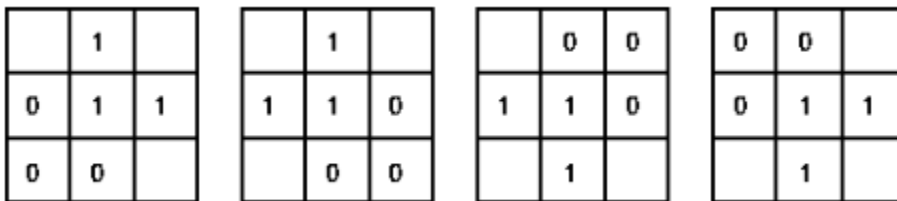
BlackHat  $I = (I \ominus B) - 1$





- Hit-and-miss transform
  - can be used to search an image for particular instances of a shape or other characteristic image feature
- Thinning
- Thickening
- Skeletonization / Medial Axis Transform

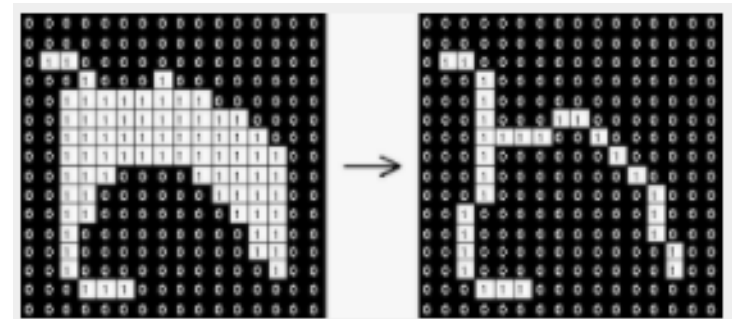
<http://homepages.inf.ed.ac.uk/rbf/HIPR2/morops.htm>



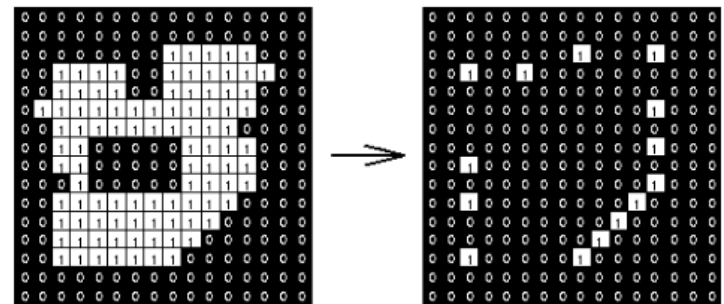
Four structuring elements used for corner finding in binary images using the **hit-and-miss transform**.

Note that they are really all the same element, but rotated by different amounts.

<https://homepages.inf.ed.ac.uk/rbf/HIPR2/hitmiss.htm>



Skeletonization by morphological thinning



Effect of the hit-and-miss based 90° angle convex corner detector on a simple binary image

- Opening
  - often used before counting regions in a binary image.
    - ◆ Ex: a thresholded image of cells on a microscope slide, to separate out cells that are near each other before counting the regions.
- Closing
  - used in most of the more sophisticated connected-component algorithms to reduce unwanted or noise-driven segments.
- Note:
  - open and close tend to preserve the area of connected regions.
- Morphological gradient
  - often used when we want to isolate the perimeters of bright regions so we can treat them as whole objects (or as whole parts of objects).
  - The complete perimeter of a region tends to be found because an expanded version is subtracted from a contracted version of the region, leaving a complete perimeter edge.

## Morphological operators usage:

### ■ Binary images

- Remove unwanted information caused by the segmentation process (ex: thresholding):
  - ◆ Noise particles
  - ◆ Holes within particles
  - ◆ Particles touching the border of an image
  - ◆ Particles touching each other
  - ◆ Particles with uneven borders
- Extract the border of objects (morphological gradient)
- Extract the skeleton of the object (and from it, endpoints and junctions)
  - ◆ can be used for recognition purposes

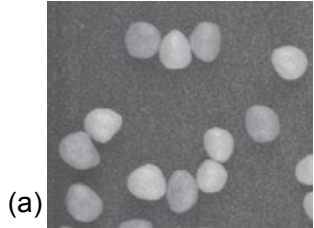
### ■ Grayscale images

- Smoothing: opening followed by a closing operation; it removes both bright and dark artifacts of noise
- Edge enhancement (morphological gradient)
- Reveal areas that are lighter (TopHat) or darker (BlackHat\*) than the surrounding region

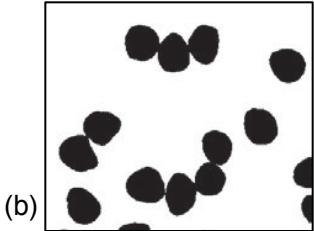
(\*) [OpenCV library name](#)

- When used correctly, morphological operations preserve the essential features of shape of an object, while removing the irrelevant details.

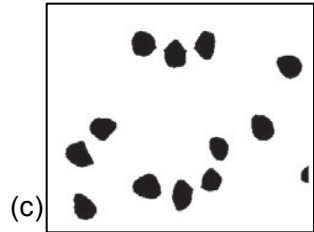
# Computer Vision Morphological operators: Binary images



(a) Image of several globular objects.



(b) Result of thresholding (a)



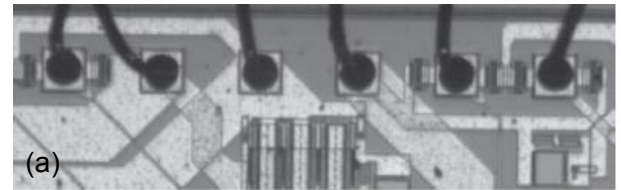
(c) Result of **eroding the inverse of (b)** with a circle of diameter 15

Note: in these 2 examples, the objects are black and the background is white

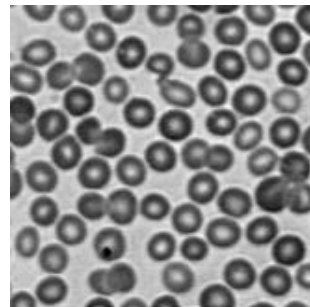
(a) Image of a ball-bonded die.  
The goal is to segment the balls.

(b) Result of thresholding (a).  
The segmentation includes the wires that are bonded to the pads.

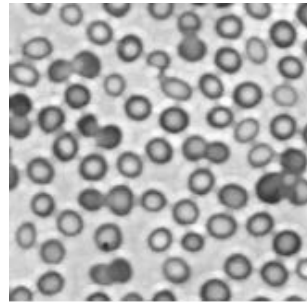
(c) Result of performing an **opening of the inverse of (b)** with a circle of diameter 31.  
The wires and the other extraneous segmentation results have been removed by the opening, and only the balls remain.



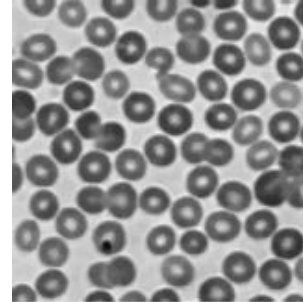
# Computer Vision Morphological operators: Gray scale images



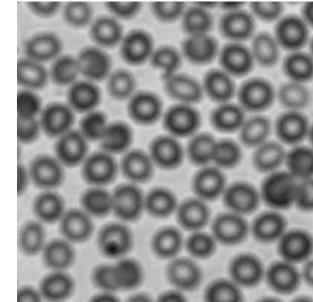
original



dilation

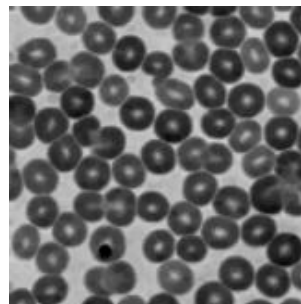


closing

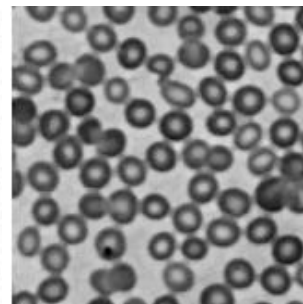


$(\text{dilation} + \text{erosion}) / 2$   
≈ smoothing

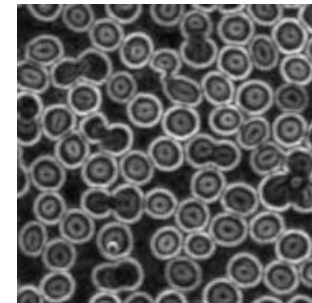
structuring element ...?



erosion

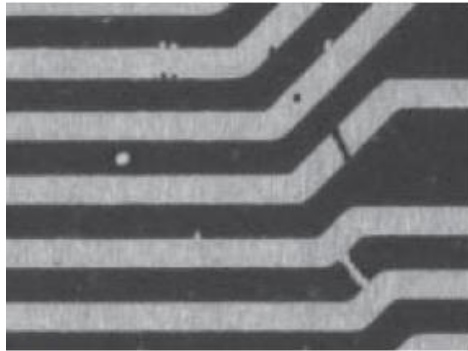


opening

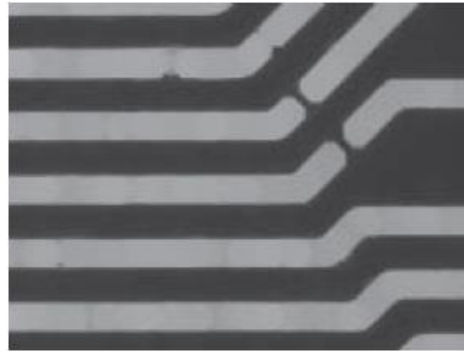


morphologic gradient

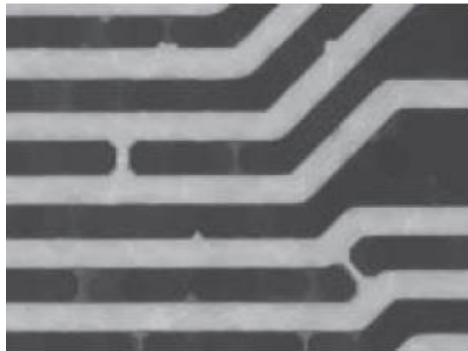
# Computer Vision Morphological operators: Gray scale images



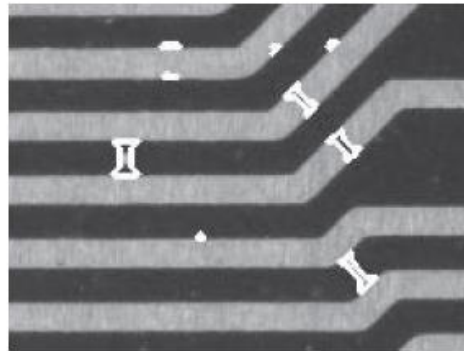
(a)



(b)



(c)



(d)

- (a) Image showing a **part of a PCB** with several tracks that have spurs, mouse bites, pinholes, spurious copper, and open and short circuits.
- (b) Result of performing a gray value opening with an octagon of diameter 11 on (a).
- (c) Result of performing a gray value closing with an octagon of diameter 11 on (a).
- (d) Result of segmenting the errors in (a) by using a dynamic threshold (\*) operation with the images of (b) and (c).

(\*) Dynamic thresholding:

is the operation of comparing the image to its local background.

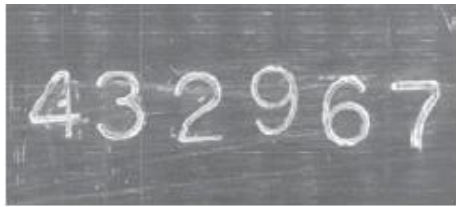
The dynamic thresholding operation for bright objects is given by

$$S = \{(r, c) \in R \mid f_{r,c} - g_{r,c} \geq \text{thr}\}$$

while the dynamic thresholding operation for dark objects is given by

$$S = \{(r, c) \in R \mid f_{r,c} - g_{r,c} \leq -\text{thr}\}$$

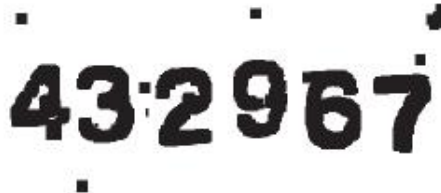
# Computer Vision Morphological operators: Gray scale images



(a)



(b)



(c)



(d)

(a) Image showing a punched serial number. Because of the scratches, texture, and illumination, it is difficult to segment the characters directly.

(b) Result of computing the morphologic gradient (difference between dilation and erosion) with a  $9 \times 9$  square.

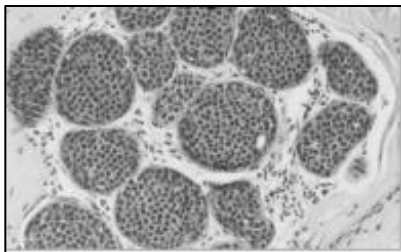
(c) Result of thresholding (b).

(d) Result of computing the connected components of (c) and selecting the characters based on their size.

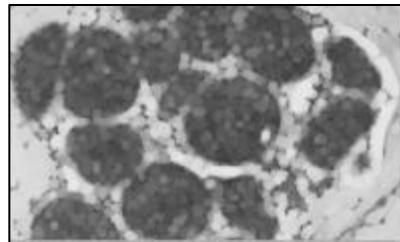
Adapted from *Handbook of Machine and Computer Vision*  
A. Hornberg (editor)



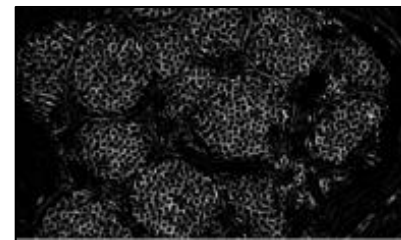
# Computer Vision Morphological operators: Gray scale images



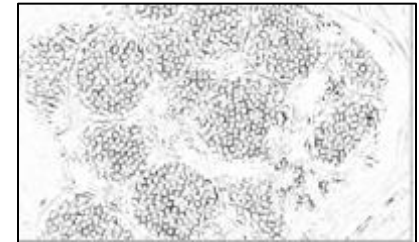
Original image



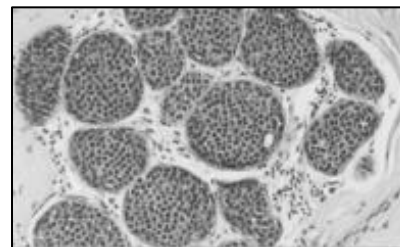
Opening



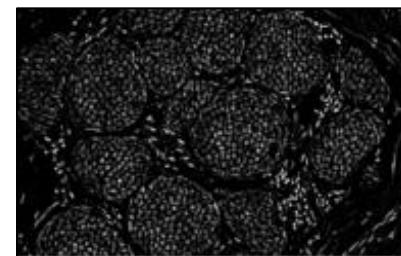
Top hat =  
 $\text{img} - \text{open}(\text{img})$



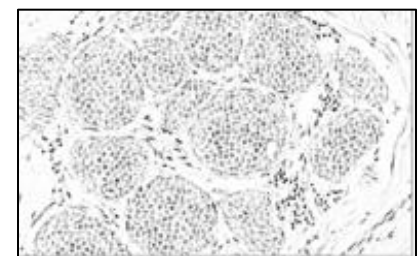
Top hat complemented



Closing



Black hat =  
 $\text{close}(\text{img}) - \text{img}$



Black hat complemented

<http://www.cyto.purdue.edu/cdroms/micro2/content/education/wirth08.pdf>